

Contrastive Learning: A Mathematical Perspective

Allowing Image And Text Data To Communicate

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CMU STAMPS Workshop: Trustworthy Statistical Inference for the Physical Sciences
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A Mathematical Perspective on Contrastive Learning

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[arXiv:2505.24134](https://arxiv.org/abs/2505.24134)

Generating Images: Prompt-to-Image Models

- ▶ Sample using Stable Diffusion XL: [Podell et al. \[4\] \(2023\)](#)
- ▶ Relies on a model to link text and images

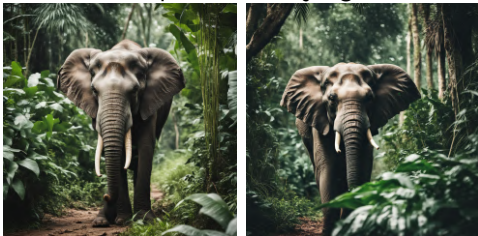
A cat and a frog



A large aircraft taking off



An elephant in the jungle



A skateboarder in California



Interpreting Images: Visual-Language Models

- ▶ Vision Question Answering analyzes visual and text data to provide responses: Wang et al. [6] (2024)
- ▶ Relies on a model to link text and images

Plant Identification



What kind of flower is this?

Model Response

The flowers in the picture are forget-me-nots (*Myosotis*). These small, delicate flowers are known for their vibrant colors, typically blue, pink, or white, and are often used in bouquets and gardens for their charming appearance.

Multilingual Text Recognition



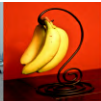
What is the text?

Model Response

The text on the wall is in Portuguese and reads: "não é normal ter medo de andar sozinha na rua!!!" This translates to: "It is not normal to be afraid to walk alone on the street!!!"

Aligning Text and Images

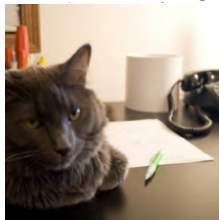
- ▶ Key contrastive learning methodology CLIP: Radford et al. [5] (2021)
- ▶ 61028 Google Scholar citations as of May 13th 2026 (44K in October)
- ▶ Cosine similarity between $a, b \in \mathbb{S}^{n_e-1}$: $\langle a, b \rangle$
- ▶ CLIP represents text as $a \in \mathbb{S}^{n_e-1}$ and image as $b \in \mathbb{S}^{n_e-1}$: then calculate $\langle a, b \rangle$.

				
The large, white jumbo jet is parked on an airport runway.	0.28	0.16	0.14	0.23
A table with a bunch of bananas hanging from a holder	0.08	0.32	0.12	0.11
Two woman on the beach holding their surfboards	0.13	0.15	0.33	0.20
The wing of an airplane flying above a beach.	0.18	0.12	0.18	0.32

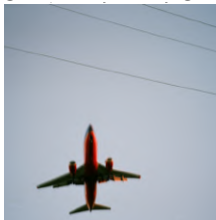
Retrieving Images Given Text MS COCO: Lin et al [3] (2014)

- ▶ 71732 Google Scholar citations as of May 13 2026
- ▶ CLIP trained on MC COCO 118K (image,text) pairs
- ▶ Retrieve closest image from 42K MS COCO test set given new test prompt

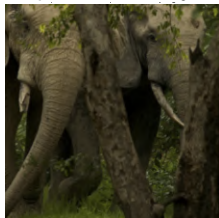
A cat and a frog



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An elephant in the jungle

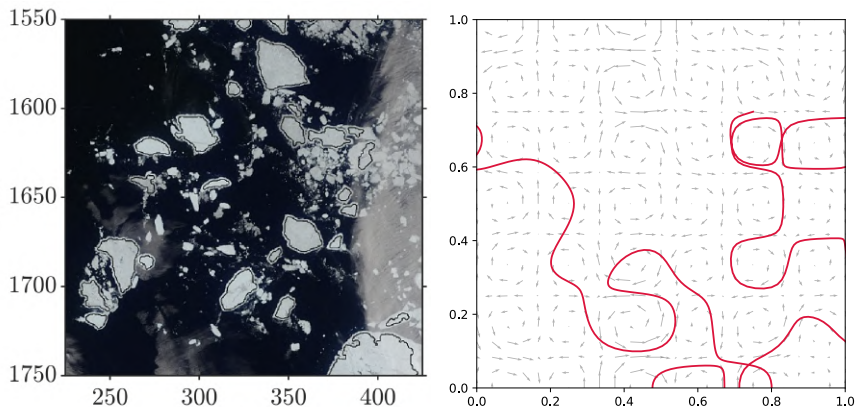


A skateboarder in California



Mapping Tracer Positions to Velocity

- ▶ Infer ocean currents conditioned on data from passive tracers (e.g., sea ice floes)
- ▶ Relies on a model relating velocity to observations



Chen et al. (2021)

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Setup for Contrastive Learning

Data is generated from a common reality

Text: $u \in \mathcal{U}$

Images: $v \in \mathcal{V}$

Distribution: $\mu(du, dv)$

Data pairs: $\{(u^i, v^i)\}_{i=1}^N \sim \mu$ i.i.d.

Embed data into a common low-dimensional space

$$g_u: \mathcal{U} \times \Theta \rightarrow \mathbb{S}^{n_e-1}$$

$$g_v: \mathcal{V} \times \Theta \rightarrow \mathbb{S}^{n_e-1}$$

Encoders are represented using data-dependent architectures:

- ▶ Text: Byte-pair encoding and transformers
- ▶ Images: Convolution layers and transformers
- ▶ L^2 normalization to map to the sphere

Contrastive Learning Problem

CLIP Objective Function Radford et al [5] (2021)

Find $\theta = (\theta_u, \theta_v, \tau)$ using only samples from μ

$$\begin{aligned} L_{\text{clip}}^N(\theta) &:= \frac{1}{N} \sum_{i=1}^N \langle g_u(u^i; \theta_u), g_v(v^i; \theta_v) \rangle / \tau \\ &\quad - \frac{1}{2N} \sum_{i=1}^N \log \left(\sum_{j=1}^N \exp(\langle g_u(u^i; \theta_u), g_v(v^j; \theta_v) \rangle / \tau) \right) \\ &\quad - \frac{1}{2N} \sum_{j=1}^N \log \left(\sum_{i=1}^N \exp(\langle g_u(u^i; \theta_u), g_v(v^j; \theta_v) \rangle / \tau) \right) \\ \theta^* &= \arg \max_{\theta} L_{\text{clip}}^N(\theta). \end{aligned}$$

Objective aligned pair samples and penalizes unaligned pairs

Population-level Picture I

B, Stuart and Tran (2025) [1]

Data Notation

Data Measure: $\mu(du, dv)$

Data Marginals: $\mu_u(du), \mu_v(dv)$

Data Conditionals: $\mu_{u|v}(du|v), \mu_{v|u}(dv|u)$

Target Notation

Target Measure: $\nu(du, dv; \theta) = \rho(u, v; \theta) \mu_u(du) \mu_v(dv)$

$$\rho(u, v; \theta) \propto \exp(\langle g_u(u; \theta_u), g_v(v; \theta_v) \rangle / \tau)$$

Target Conditionals: $\nu_{u|v}(du|v; \theta), \nu_{v|u}(dv|u; \theta)$

Population-level Picture II

B, Stuart and Tran (2025) [1]

Objective Function: Population

$$J(\theta) = \frac{1}{2} \mathbb{E}_{v \sim \mu_v} [\text{D}_{\text{kl}}(\mu_{u|v} || \nu_{u|v}(\cdot; \theta))] + \frac{1}{2} \mathbb{E}_{u \sim \mu_u} [\text{D}_{\text{kl}}(\mu_{v|u} || \nu_{v|u}(\cdot; \theta))]$$
$$\theta^* = \arg \min_{\theta} J(\theta)$$

The non-parametric minimizer is $\nu_{u|v}(\cdot; \theta) = \mu_{u|v}$ and $\nu_{v|u}(\cdot; \theta) = \mu_{v|u}$.

Objective Function: Empirical

$$J(\theta) = \text{const} - L(\theta)$$
$$L(\theta) = \mathbb{E}_{(u,v) \sim \mu} \langle g_u(u; \theta_u), g_v(v; \theta_v) \rangle / \tau$$
$$- \frac{1}{2} \mathbb{E}_{v \sim \mu_v} \log \mathbb{E}_{u' \sim \mu_u} \exp(\langle g_u(u'; \theta_u), g_v(v; \theta_v) \rangle / \tau)$$
$$- \frac{1}{2} \mathbb{E}_{u \sim \mu_u} \log \mathbb{E}_{v' \sim \mu_v} \exp(\langle g_u(u; \theta_u), g_v(v'; \theta_v) \rangle / \tau).$$
$$\theta^* = \arg \max_{\theta} L(\theta)$$

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Analysis of CLIP: Gaussian Setting B, Stuart and Tran (2025) [1]

Goal: Analyze closed-form solutions for CLIP optimization

Setting: Gaussian data distribution $\mu = \mathcal{N}(0, \mathcal{C})$ with block covariance matrix

$$\mathcal{C} = \begin{bmatrix} \mathcal{C}_{uu} & \mathcal{C}_{uv} \\ \mathcal{C}_{vu} & \mathcal{C}_{vv} \end{bmatrix}$$

Linear Model: Project $u \in \mathbb{R}^{n_u}$ and $v \in \mathbb{R}^{n_v}$ into $\mathbb{R}^{n_e}$

$$\begin{aligned} g_u(u) &= Gu, & G &\in \mathbb{R}^{n_u \times n_e}, \\ g_v(v) &= Hv & H &\in \mathbb{R}^{n_v \times n_e}. \end{aligned}$$

CLIP Model:

$$\begin{aligned} \nu(u, v; \theta) &\propto \exp(\langle Gu, Hv \rangle) \mu_u(u) \mu_v(v) \\ &= \exp(\langle u, Av \rangle) \mu_u(u) \mu_v(v), \quad \theta = A = G^\top H. \end{aligned}$$

Analysis of CLIP: Gaussian Setting B, Stuart and Tran [1]

Theorem: No Dimension Reduction

Minimizing J over A with $n_e = \min(n_u, n_v)$ has solution

$$A = C_{uu}^{-1} C_{uv} C_{vv}^{-1}$$

results in the conditional distributions

$$\nu_{u|v}(u|v; \theta^*) = \mathcal{N}(C_{uv} C_{vv}^{-1} v; C_{uu})$$

$$\nu_{v|u}(v|u; \theta^*) = \mathcal{N}(C_{vu} C_{uu}^{-1} u; C_{vv})$$

Corollary: Conditional means of $\mu_{u|v}, \mu_{v|u}$ are recovered, but not variances

Theorem: With Dimension Reduction

Minimizing J over A with $n_e < \min(n_u, n_v)$ has solution

$$A = C_{uu}^{-\frac{1}{2}} (C_{uu}^{-\frac{1}{2}} C_{uv} C_{vv}^{-\frac{1}{2}})_{n_e} C_{vv}^{-\frac{1}{2}}$$

Corollary: Small n_e yields low rank approximations of the conditional mean

Analysis of CLIP: Novel Objective Functions

$J_{\text{one-sided}}$, J_{joint} (variants on J) introduced in [B, Stuart and Tran \(2025\)](#) [1]

Original: Match Both Conditionals

$$J(\theta) = \frac{1}{2} \mathbb{E}_{v \sim \mu_v} [D_{\text{kl}}(\mu_{u|v} \| \nu_{u|v}(\cdot; \theta))] + \frac{1}{2} \mathbb{E}_{u \sim \mu_u} [D_{\text{kl}}(\mu_{v|u} \| \nu_{v|u}(\cdot; \theta))].$$

Variant: Match One Conditional

$$J_{\text{one-sided}}(\theta) = \mathbb{E}^{v \sim \mu_v} D_{\text{kl}}(\mu_{u|v} \| \nu_{u|v}(\cdot; \theta)).$$

Variant: Match Joint

$$J_{\text{joint}}(\theta) = D_{\text{kl}}(\mu \| \nu(\cdot; \theta)).$$

Analysis of CLIP: Novel Objective Functions

Theorem 1

When embedding dimension is large enough:

- ▶ minimizing J exactly matches conditional means; overestimates conditional covariances
- ▶ minimizing $J_{\text{one-sided}}$ exactly matches the conditional mean and covariance

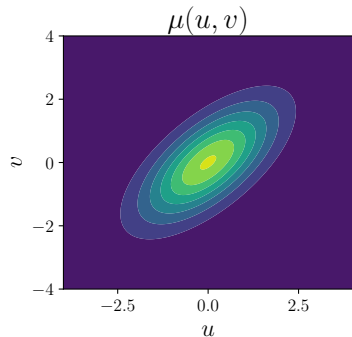
Theorem 2

When embedding dimension is large enough:

- ▶ minimizing J underestimates marginal covariances
- ▶ minimizing J_{joint} more closely estimates marginal covariances

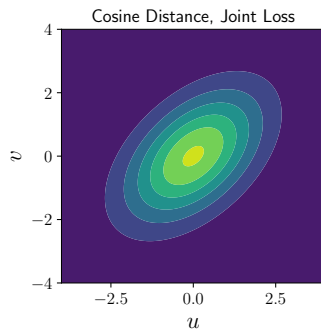
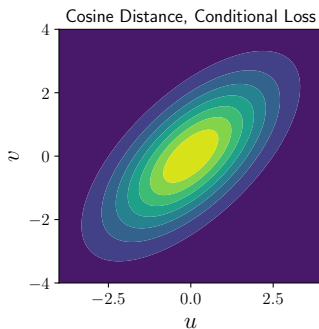
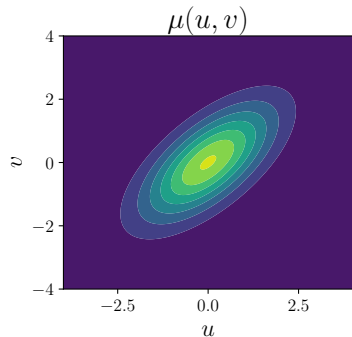
Visualizations of CLIP Generalizations I

- ▶ Two-dimensional Gaussian data distribution $\mu = \mathcal{N}(0, \mathcal{C})$
- ▶ Used un-normalized linear encoders to preserve Gaussian structure



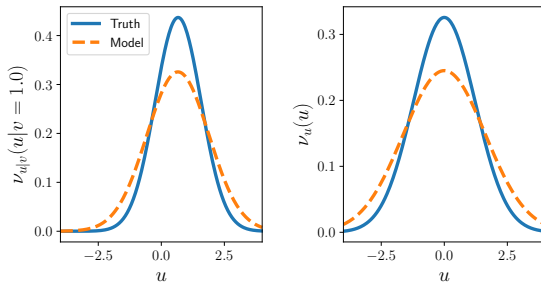
Visualizations of CLIP Generalizations I

- ▶ Two-dimensional Gaussian data distribution $\mu = \mathcal{N}(0, \mathcal{C})$
- ▶ Used un-normalized linear encoders to preserve Gaussian structure



Visualizations of CLIP Generalizations II

CLIP with two-sided conditional loss



CLIP with joint loss

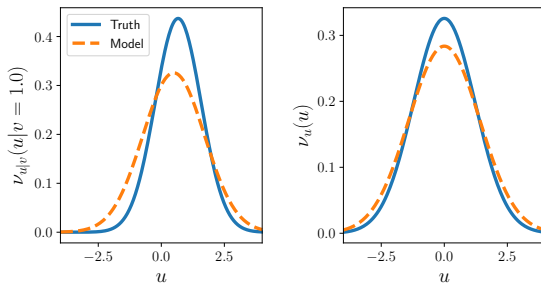


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CLIP Learning Problem

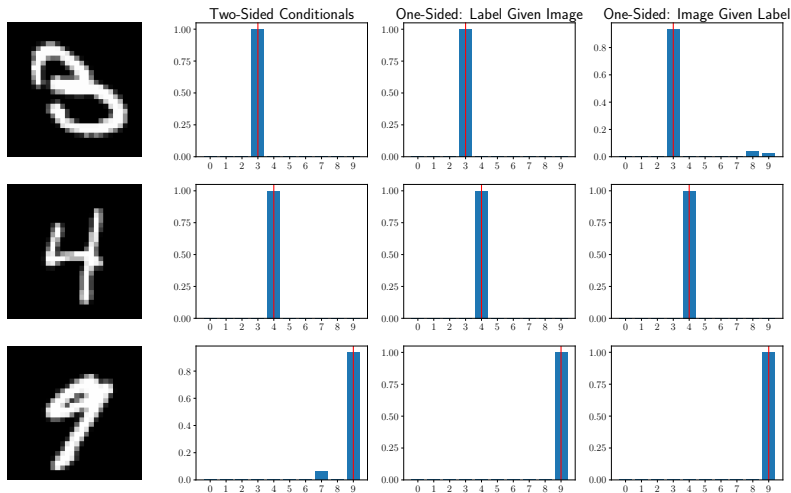
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Application: MNIST

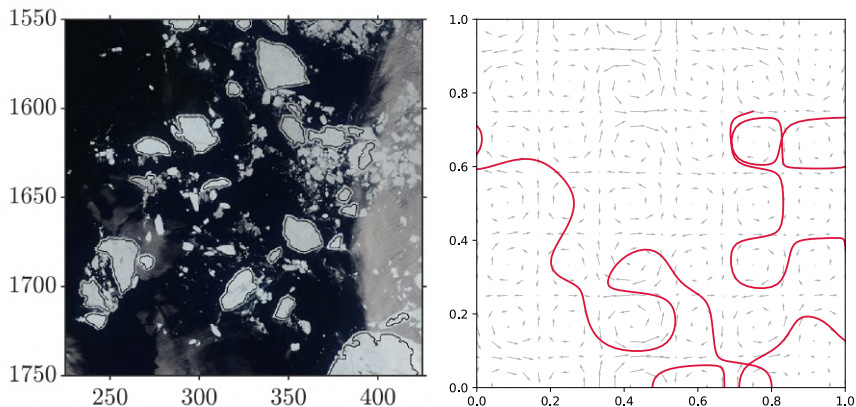
- ▶ Data modalities: images $u \in \mathcal{U} = [0, 1]^{28 \times 28}$ and digits $v \in \mathcal{V} = \{0, \dots, 9\}$
- ▶ Performed classifications using models learned with different loss functions



Application: Lagrangian Data Assimilation

Original problem formulation: Kuznetsov, Ide and Jones (2003) [2]

- ▶ Infer ocean currents conditioned on data from passive tracers (e.g., sea ice floes)
- ▶ Typically relies on a model relating velocity to observations

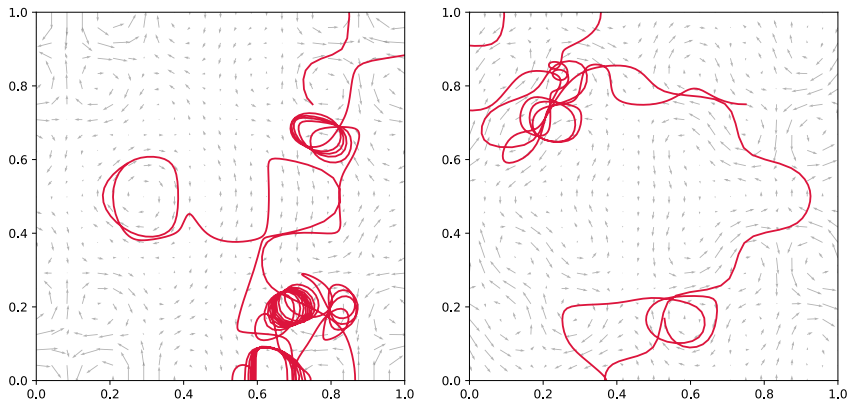


Chen et al. (2021)

Application: Lagrangian Data Assimilation

- ▶ Image: Velocity Field is represented using a time-dependent potential
- ▶ Time-series: Lagrangian trajectory of a tracer's position
- ▶ Velocity is encoded using Fourier representation of the potential

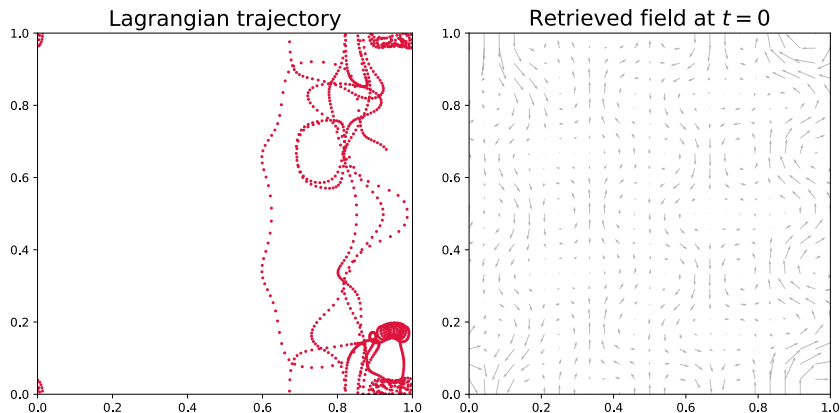
Paired potentials (background) and trajectories (red)



Application: Lagrangian Data Assimilation

- ▶ CLIP model aligns trajectory and time-dependent potentials
- ▶ Retrieval identifies the most likely velocity from a database given a trajectory

Paired potentials (background) and trajectories (red)



Application: Lagrangian Data Assimilation

- Measured accuracy of retrieval between both modalities

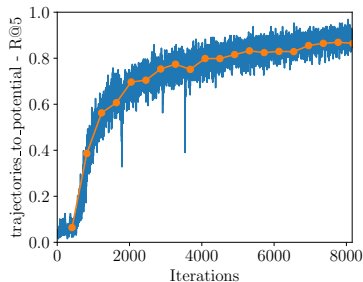
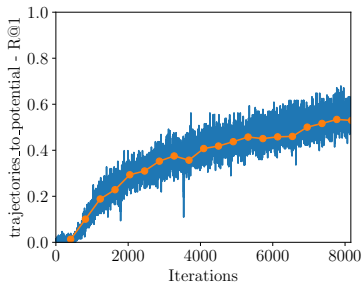
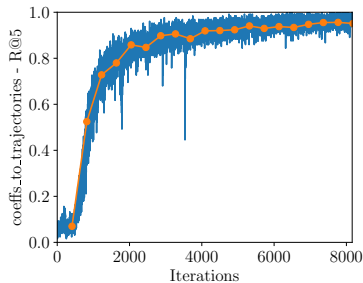
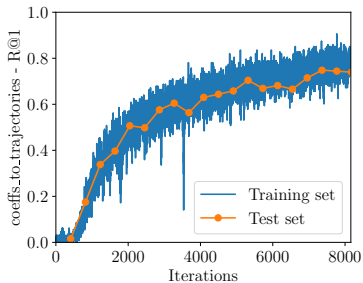


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Main Messages

- ▶ Contrastive learning relates two data modalities by tilting a product distribution
- ▶ Theory in Gaussian setting:
 - ▶ Standard loss matches both conditional means
 - ▶ Connections to low rank matrix approximation
- ▶ Application to Lagrangian data assimilation

Outlook

- ▶ Extending framework to more than two modalities
- ▶ Theoretical study of compositional generalization
- ▶ Other applications in science and engineering

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